

RENE LEDESMA

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Experience

VC Extern, EIC

May 2026 – Jun 2026

Software Engineer Extern

Remote

- Conducted due diligence on potential investment opportunities, analyzing company data such as fundraising history, team composition, press coverage, and technology.
- Leveraged AI tools for market research, creating reports on market size, industry trends, and growth potential to guide strategic investment decisions.
- Produced synthesis reports on key areas, including market opportunity, technology, competitive landscape, team assessment, and risk analysis, culminating in a professional investment memo.

OS Labs - Developer Caching Utility

Mar 2026 – May 2026

Software Engineer

New York, NY

- Engineered a robust cache invalidation utility utilizing mutual exclusion (mutex) request locking and probabilistic early-expiration, mitigating thundering herd bottlenecks and increasing API throughput by 45%.
- Optimized memory management by designing a discrete UUID tagging system for cached payloads, reducing server memory footprint by 30% and eliminating the risk of serving stale client data during TTL expirations.
- Streamlined data conversion pipelines by rigorously balancing algorithmic time complexity with resource constraints, reducing average data retrieval latency by 50ms under peak concurrent loads.

Boltcity

May 2025 – Jun 2025

IT Infrastructure Technician

Manhasset, NY

- Executed the end-to-end relocation and configuration of luxury retail IT infrastructure, successfully migrating POS terminals, VoIP networks, and A/V systems with zero operational downtime.
- Accelerated project completion by 2 days ahead of schedule by utilizing Linux command-line tools to troubleshoot IP ports and rapidly deploy credit card routing infrastructure.

New York City Department of Education

Apr 2015 – Jun 2023

Paraprofessional / SPOC for Technology

New York, NY

- Directed technology management and resolved core infrastructure bottlenecks for a public elementary school, increasing educational application uptime by 30%.

Projects

Golden Quill | *TypeScript, React, Node.js, Express, Pinecone, OpenAI, LangChain*

- Architected a highly scalable RAG application integrating OpenAI's LLM and a Pinecone vector database, reducing creative workflow blockers by delivering context-aware generations with sub-800ms latency.
- Implemented automated MLOps pipelines using Winston for logging and custom precision/recall evaluation metrics, reducing production model drift and improving response accuracy by 25%.

Trading Bot | *Python, C, C++, JavaScript, Docker*

- Architected a high-frequency trading bot across a polyglot stack (C, C++, Python, JavaScript), engineering a custom in-memory caching suite that bypassed disk I/O bottlenecks and reduced function execution latency to 0.02ms.
- Implemented a continuous algorithmic feedback loop to dynamically adjust signal weights based on market behavior, utilizing a custom caching architecture to ensure real-time optimization with sub-0.01ms latency.

Automated Continuous Deployment Workflow | *Docker, CI/CD, GitHub Actions, Node.js*

- Architected a robust CI/CD pipeline utilizing GitHub Actions and Docker, automating the testing and deployment lifecycle to accelerate continuous feature delivery by 40%.
- Containerized application environments to ensure strict parity across staging and production infrastructure, eliminating manual configuration drift and reducing deployment failure rates by 25%.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, HTML, CSS/Sass, SQL, Python

Frameworks: React (Router, Hooks, Context, Recoil), Redux, Node.js, Express.js, PostgreSQL, MongoDB, Mongoose, Jest, Django

Developer Tools: Git, Docker, Webpack, Vite, Microsoft Azure, CI/CD, GitHub Actions, JIRA, Grafana

Education

CUNY - John Jay College of Criminal Justice

Aug 2022 – May 2026

Bachelor of Science in Criminal Justice

New York, NY

Codesmith

Jul 2025 – May 2026

Advanced Residency for Software Engineering

New York, NY

PC AGE Career Institute

Dec 2024 – Nov 2025

Internetwork Engineering Certificate

Jersey City, NJ